

University of London
Imperial College of Science, Technology and Medicine
Department of Physics

Ciao, I'm Fabio, I Am Very Interested in what Makes Topology Work

Peru d'Ornellas

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Abstract

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Contents

1	Introduction	7
1.1	Go on a different lattice	7
1.2	Quantum Hall and Topological Insulators	7
1.2.1	Initial motivation - The quantum hall effect	7
1.2.2	The TKNN Formula and connection to Chern number	7
1.3	Spin Liquids	7
1.3.1	background - anderson RVB, frustrated magnets, X-G Wen and co	7
1.3.2	Kitaev Model	7
1.3.3	Physical materials	7
1.4	Thesis Outline	7
2	Lattices and Their Transformations	8
2.1	Voronoi Lattices	8
2.2	LLoyd's Algorithm	8
2.3	Dual Lattices	8
2.4	Vertex Expansion	8
3	Chern Number and Topological Insulators	9
3.1	Abelian Berry Phase and the Chern Number	10
3.1.1	The Simplest Story – A Discrete Lattice of States	10
3.1.2	Topology of a General Hamiltonian	13
3.1.3	An Example – The Two-Level System	16
3.1.4	Final Notes	19
3.2	The Qi-Wu-Zhang Model	20
3.2.1	Strip Geometry and Edge Modes	21
3.2.2	The Amorphous QWZ Model	23
3.2.3	Topology of the Amorphous QWZ Model	25
	References	25

4	Local Markers	27
4.1	Kitaev's Local Marker	28
4.1.1	Relation to the Chern Number	30
4.1.2	Kitaev's Marker in a Finite System	33
4.2	The Bott Index	33
4.3	Chern Marker	36
	References	37
5	The Crosshair Marker	40
6	Kitaev's Honeycomb Model	41
7	The Amorphous Kitaev Model	42
8	The Star Lattice Kitaev-Heisenberg Model	43
9	Conclusion and Outlook	44

Chapter 1

Introduction

1.1 Go on a different lattice

1.2 Quantum Hall and Topological Insulators

1.2.1 Initial motivation - The quantum hall effect

1.2.2 The TKNN Formula and connection to Chern number

1.3 Spin Liquids

1.3.1 background - anderson RVB, frustrated magnets, X-G Wen and co

1.3.2 Kitaev Model

1.3.3 Physical materials

1.4 Thesis Outline

Chapter 2

Lattices and Their Transformations

2.1 Voronoi Lattices

2.2 LLoyd's Algorithm

2.3 Dual Lattices

2.4 Vertex Expansion

Chapter 3

Chern Number and Topological Insulators

Paragraph introducing the topics covered in this section. Points to hit:

- Periodic translational hamiltonians can be classified according to the chern number of their bands, where nonzer chern number implies the existence of gapless edge modes in open boundaries.
- This classification has been immensely successful for understanding a lot of materials eg: QHE, Spin Hall effect etc
- The classification relies on a notion of momentum space - in order to construct a manifold of states on which to define parallel transport.
- Thus it doesn't work for systems without k space - eg. open boundaries/disorder
- Local markers are an attempt at constructing a real space version of this
- We will explain all the prerequisites for constructing a local marker, as well as discussing their limitations.

Throughout this chapter and the next we will focus on non-interacting electrons on a lattice, thus let us begin by defining our Hilbert space in a first-quantised framework. We will work with the most general definition of a non-interacting tight-binding Hamiltonian in two dimensions. The Hilbert space $\mathcal{H}$ contains a set of N sites, where site j is at a position $\mathbf{r}_j$, and has η internal degrees of freedom. For example, our sites could correspond to atoms, and the states at each site could represent individual orbitals around an atom. We will not assume that our sites are arranged in a regular lattice – we will also want to model amorphous structures – thus, a state in this system can be written as

$$|\psi\rangle = \sum_j c_{\mathbf{r}_j, \alpha} |\mathbf{r}_j\rangle \otimes |\alpha\rangle, \quad (3.1)$$

where $c_{\mathbf{r}_j, \alpha}$ are a set of complex coefficients and $\alpha \in \{1 \dots \eta\}$ labels the internal degree of freedom. A general Hamiltonian in this context takes the form

$$H = \sum_{j, k, \alpha, \beta} H_{j, k, \alpha, \beta} |\mathbf{r}_j\rangle \langle \mathbf{r}_k| \otimes |\alpha\rangle \langle \beta|. \quad (3.2)$$

If the system has translational symmetry $\{\mathbf{r}_j\}$ will define a regular lattice with periodic boundaries and the Hamiltonian is invariant under translations – we can of course diagonalise the Hamiltonian in a basis of k -states,

$$|\mathbf{k}\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{r}_j} e^{-i\mathbf{k}\mathbf{r}_j} |\mathbf{r}_j\rangle. \quad (3.3)$$

For a finite system the k -states are quantised to $k_a = 2\pi n/L_a$, where L_a is the length of the system in direction $a \in \{x, y\}$, whereas in the case of an infinite system k can take any value in the interval $[0, 2\pi]$.

3.1 Abelian Berry Phase and the Chern Number

In this section we introduce the mathematical construction of Berry phase, Berry flux and Chern number. We will present these ideas in two different contexts. In §3.1.1 we introduce the simplest possible story that captures the essentials of Berry phase and Chern number: that of a discretised lattice of quantum states. Here we will miss many of the details that make such concepts relevant to physical systems, however the picture is sufficiently simple as to provide a practical intuition for how to calculate these quantities. In §3.1.2, we will do things *properly*, by considering the effect on the band structure of some Hamiltonian that is changed adiabatically over some smooth parameter space. We will derive the exact form of the Chern number in this context and relate it to the TKNN invariant [1]. Finally, in §3.1.3 we give a brief example of such a calculation, demonstrating how these concepts apply to a two-state quantum system.

3.1.1 The Simplest Story – A Discrete Lattice of States

In quantum mechanics a state is defined as an equivalence class of vectors, where two states that differ only by a complex phase are considered physically equivalent,

$$|\psi\rangle \sim e^{-i\alpha} |\psi\rangle. \quad (3.4)$$

Thus, we seem to have a degeneracy built into our description of physics. This means that one can define a gauge transformation – multiplying a state with a complex phase – that is guaranteed not to affect the system in any ‘real’ measurable way. Any derived quantity must be invariant under gauge transformations in order to be considered physical. Thus, one would expect any quantity that depends on the state’s phase to be unphysical.

Similarly, considering some arbitrary set of states in the same Hilbert space, $\{|\psi_j\rangle\}$, we may define a local gauge transformation where we apply a different phase to each individual state,

$$|\psi_j\rangle \rightarrow e^{-i\alpha_j} |\psi_j\rangle. \quad (3.5)$$

Again, the physical meaning of each state is unaffected, so we expect that any physical quantity calculated from these states must be invariant under the transformation. Thus one might expect that any *phase-like* quantity calculated from these states should have no physical relevance. For

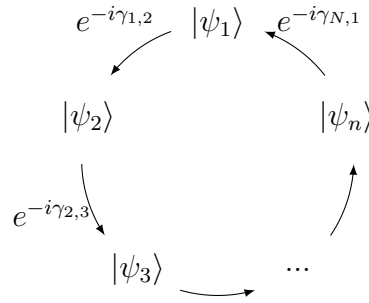
example, let us consider the relative phase between state j and k , $\gamma_{j,k}$, defined according to

$$e^{-i\gamma_{j,k}} = \frac{\langle \psi_j | \psi_k \rangle}{|\langle \psi_j | \psi_k \rangle|}. \quad (3.6)$$

Clearly this quantity is invariant if all states pick up the same phase, however it is not invariant under a local transformation, changing as

$$e^{-i\gamma_{j,k}} \rightarrow e^{-i(\gamma_{j,k} + \alpha_j - \alpha_k)}, \quad (3.7)$$

and so cannot be a physical observable. In order to construct a gauge-invariant object, let's pick a set of n (≥ 3) states $|\psi_j\rangle$ and arrange them in a loop.



We can now define a new quantity called the Berry phase:

$$e^{-i\gamma_B} = \frac{\langle \psi_1 | \psi_2 \rangle \langle \psi_2 | \psi_3 \rangle \dots \langle \psi_N | \psi_1 \rangle}{|\langle \psi_1 | \psi_2 \rangle \langle \psi_2 | \psi_3 \rangle \dots \langle \psi_N | \psi_1 \rangle|} \quad (3.8)$$

Another way of writing this would be

$$\gamma_B = -\arg \text{Tr} [\hat{\psi}_1 \hat{\psi}_2 \dots \hat{\psi}_N], \quad (3.9)$$

where the trace is over the whole system, and $\hat{\psi}$ is shorthand for the density operator

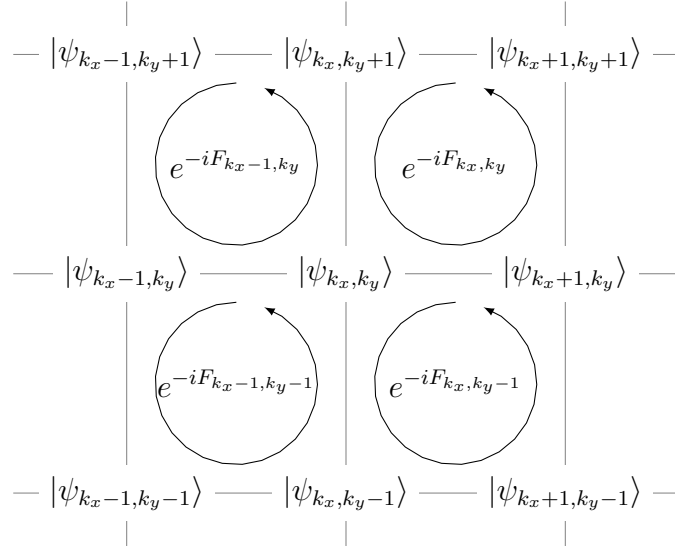
$$\hat{\psi}_i = |\psi_i\rangle \langle \psi_i|. \quad (3.10)$$

This quantity is clearly gauge-invariant, since any phase applied to a state $|\psi_m\rangle$ will show up twice, once from $\langle \psi_m | \psi_{m+1} \rangle$ and once from $\langle \psi_{m-1} | \psi_m \rangle$. The phase will have opposite sign in each of these expressions so they cancel out. Furthermore, we can see that the Berry phase around a loop is simply the sum of all the relative phases around that loop modulo 2π .

$$\gamma_B = \sum_{n=1}^N \gamma_{n,n+1} \mod 2\pi \quad (3.11)$$

Berry Flux

Now suppose that we have a large number of states arranged in a 2D grid. These states $|\psi_{\mathbf{k}}\rangle$ are labelled according with a pair of indices k_x and k_y . In general, this parameter $\mathbf{k}$ will refer to the crystal momentum, however the construction is equally valid for any other parameter that defines a 2D grid of quantum states.



We introduce a new quantity called the Berry flux. This is defined as the Berry phase around a plaquette on our grid,

$$F_{\mathbf{k}} = -\arg \text{Tr} \left[\hat{\psi}_{k_x, k_y} \hat{\psi}_{k_x+1, k_y} \hat{\psi}_{k_x+1, k_y+1} \hat{\psi}_{k_x, k_y+1} \right]. \quad (3.12)$$

$F_{\mathbf{k}}$ is just the sum (mod 2π) of the relative phases along each edge of the plaquette starting at point $\mathbf{k}$. Thus, the Berry phase around some arbitrary loop containing multiple plaquettes can be found by summing the value of $F_{\mathbf{k}}$ for every plaquette contained in that region. Each relative phase flips sign depending on which way round it is evaluated. Any internal edge is shared between two plaquettes and so will appear twice in the sum of Berry fluxes, with opposite signs. This means all internal edges have their contribution to the total phase cancelled out and we are left with the sum of the relative phases only around the boundary of our region.

$$\exp \left[-i \sum_{\text{inside region}} F \right] = \exp \left[-i \sum_{\text{boundary}} \gamma \right] \quad (3.13)$$

This resembles Stokes' law, however there is a subtlety worth mentioning. All we have required is that the complex exponential of each sum is the same. This does **not** mean that the sum of the Berry flux inside a region is equal to the sum of relative phases around the boundary, since they are allowed to differ by an arbitrary multiple of 2π .

add a touch more here - it was obvious to k but maybe not to everyone

Chern Number

Now let's assume that our grid of $\mathbf{k}$ points has periodic boundaries, with $1 \leq k_x \leq N_x$ and $1 \leq k_y \leq N_y$. That is, the lattice is a discretisation of the torus – a closed manifold. The product of all the Berry flux phases is now always equal to 1,

$$\prod_{k_x=1}^{N_x} \prod_{k_y=1}^{N_y} e^{-iF_{k_x, k_y}} = 1. \quad (3.14)$$

Here because every edge in our grid is now an internal edge and thus they cancel out. Again, as before, this does not mean that the sum of all Berry fluxes in our system vanishes. It means that the sum must always equal some multiple of 2π . This sum gives us the Chern number, $\mathcal{Q}$,

$$\mathcal{Q} = \frac{1}{2\pi} \sum_{k_x, k_y} F_{k_x, k_y}. \quad (3.15)$$

$\mathcal{Q}$ is necessarily restricted to integer values, since a non-integer value would contradict equation 3.14.

Maybe its more consistent to use u rather than ψ here

3.1.2 Topology of a General Hamiltonian

Now that we have a basic intuition for what the Chern number is, let's see how this applies to a real physical system. We start by defining some arbitrary Hamiltonian $H(\mathbf{R})$, which depends on some smooth parameter $\mathbf{R} \in \mathcal{R}$.

For every value of $\mathbf{R}$, the Hamiltonian will generally have some set of eigenstates $|\psi_n(\mathbf{R})\rangle$ with corresponding energies ε_n . As before, these eigenstates are only strictly defined up to a phase. Thus let consider a single state $|\psi(\mathbf{R})\rangle$ and define a local gauge transformation as applying some $\mathbf{R}$ -dependent phase $\alpha(\mathbf{R})$ to all of the states over this band.

$$|\psi(\mathbf{R})\rangle \rightarrow e^{-i\alpha(\mathbf{R})} |\psi(\mathbf{R})\rangle. \quad (3.16)$$

In order for this to be well-behaved we will need to use a *smooth gauge*. This is a choice of gauge that ensures that the $\mathbf{R}$ -derivative of a state, $\nabla_{\mathbf{R}} |\psi(\mathbf{R})\rangle$ is always well-defined – that is, $\alpha(\mathbf{R})$ does not jump around discontinuously. In general it is not always possible to pick a gauge that is globally smooth over all of $\mathcal{R}$, however it *is* always possible to pick a locally smooth gauge around the neighbourhood of some point in $\mathcal{R}$.

Let us define relative phase between a state at $\mathbf{R}$ and another state that is infinitesimally displaced from it, at $\mathbf{R} + d\mathbf{R}$,

$$e^{-i\Delta\gamma} = \frac{\langle\psi(\mathbf{R})|\psi(\mathbf{R} + d\mathbf{R})\rangle}{|\langle\psi(\mathbf{R})|\psi(\mathbf{R} + d\mathbf{R})\rangle|}. \quad (3.17)$$

In the limit of small $d\mathbf{R}$ we can express $|\psi(\mathbf{R} + d\mathbf{R})\rangle$ to first order in $d\mathbf{R}$ as

$$|\psi(\mathbf{R} + d\mathbf{R})\rangle = |\psi(\mathbf{R})\rangle + d\mathbf{R} \cdot \nabla_{\mathbf{R}} |\psi(\mathbf{R})\rangle. \quad (3.18)$$

Which means that

$$\langle\psi(\mathbf{R})|\psi(\mathbf{R} + d\mathbf{R})\rangle = 1 + \langle\psi(\mathbf{R})|\nabla_{\mathbf{R}}|\psi(\mathbf{R})\rangle \cdot d\mathbf{R}. \quad (3.19)$$

Now, noticing that $\langle\psi(\mathbf{R})|\nabla_{\mathbf{R}}|\psi(\mathbf{R})\rangle$ is pure imaginary, we can show that

$$\Delta\gamma = \mathbf{A} \cdot d\mathbf{R}, \quad (3.20)$$

$$\mathbf{A}(\mathbf{R}) = i \langle\psi(\mathbf{R})|\nabla_{\mathbf{R}}|\psi(\mathbf{R})\rangle. \quad (3.21)$$

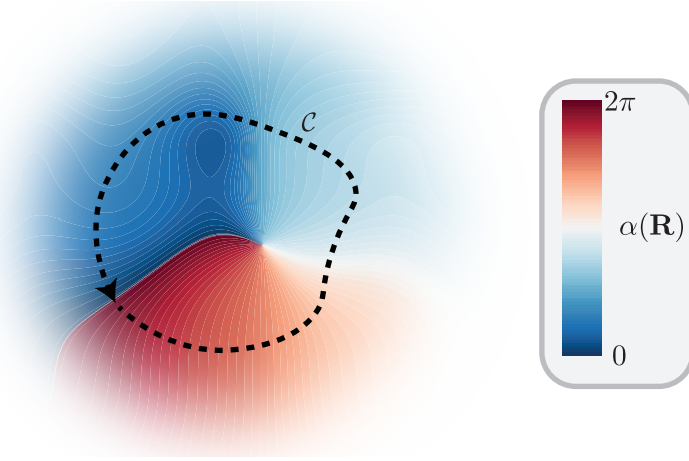


Figure 3.1: An example of a non-smooth gauge transformation designed to change the value of the Berry phase around the path $\mathcal{C}$ – marked with a dashed line – by an addition of 2π . Only a small region of $\mathbf{R}$ -space is shown, with the local phase applied shown by the colormap.

The quantity multiplying $d\mathbf{R}$ is the Berry connection $\mathbf{A}(\mathbf{R})$. Like the relative phase in the previous section, it is not a gauge-invariant quantity. Under a gauge transformation it changes according to

$$\mathbf{A}(\mathbf{R}) \rightarrow \mathbf{A}(\mathbf{R}) - \nabla_{\mathbf{R}}\alpha(\mathbf{R}) \quad (3.22)$$

Now let us suppose that we slowly transform our Hamiltonian around some closed path $\mathcal{C}$ in $\mathbf{R}$ -space and keep track of the phase picked up by a given state.¹ In analogy with eqn. 3.11, the Berry connection allows us to calculate the total phase accumulated over this route, according to

$$\gamma(\mathcal{C}) = \oint_{\mathcal{C}} \mathbf{A} \cdot d\mathbf{R}. \quad (3.23)$$

The exponential of the phase $e^{-i\gamma(\mathcal{C})}$ around the loop is gauge invariant, one can see this intuitively by realising that it is identical to the Berry phase from the discrete case, but in the limit where the discrete steps become arbitrarily close together. However, the Berry phase $\gamma(\mathcal{C})$ itself is not strictly gauge invariant, since one could imagine introducing a gauge transformation that introduces a branch cut around the path $\mathcal{C}$. After traversing the whole path, the state at the end of the loop would have picked up a total phase 2π , meaning that under a gauge transformation we can change the berry phase according to

$$\gamma(\mathcal{C}) \rightarrow \gamma(\mathcal{C}) + 2\pi n \text{ for } n \in \mathbb{Z}. \quad (3.24)$$

¹Many sources (e.g. [2]) present this picture as an *adiabatic* transformation, where one has to do a little extra work to keep track of the energy-dependent phase that a state picks up $\sim e^{-i\varepsilon t}$ alongside the topological phase determined by $\mathbf{A}$. We won't bother, and instead just consider comparing the *instantaneous eigenstates* at different values of $\mathbf{R}$.

Berry Curvature

Let us now define a quantity equivalent to the Berry flux from the previous discussion. Starting with eqn. 3.23, we use Stokes' theorem to transform it into a surface integral over a surface $\mathcal{S}$ bounded by the curve $\mathcal{C}$. We will consider the case for a two and three dimensional parameter space $\mathcal{R}$, but of course this argument generalises to arbitrary dimensions.

$$\text{In 2D: } \gamma(\mathcal{C}) = \int_{\mathcal{S}} d^2 R (\partial_x A_y(\mathbf{R}) - \partial_y A_x(\mathbf{R})), \quad (3.25)$$

$$\text{In 3D: } \gamma(\mathcal{C}) = \int_{\mathcal{S}} d\mathbf{S} \cdot (\nabla_{\mathbf{R}} \times \mathbf{A}(\mathbf{R})), \quad (3.26)$$

where we use the shorthand $\partial_{\alpha} = \frac{\partial}{\partial R_{\alpha}}$. Thus we may define the Berry curvatures,

$$\Omega(\mathbf{R}) = \partial_x A_y(\mathbf{k}) - \partial_y A_x(\mathbf{R}) \quad (3.27)$$

$$\mathbf{\Omega}(\mathbf{R}) = \nabla_{\mathbf{R}} \times \mathbf{A}(\mathbf{R}). \quad (3.28)$$

Two Useful Ways to Re-express the Berry Curvature — Now let us focus only on the 2D case and reformulate this expression in two different ways.

1. The first way finds the curvature in terms of the rate of change of the Hamiltonian only. Writing this quantity out explicitly in terms of the states $|\psi(\mathbf{R})\rangle$, we arrive at

$$\Omega(\mathbf{R}) = i \langle \partial_x \psi(\mathbf{R}) | \partial_y \psi(\mathbf{R}) \rangle + h.c. \quad (3.29)$$

We can insert a resolution of the identity $\mathbb{1} = \sum_n |\psi_n\rangle \langle \psi_n|$ to obtain

$$\Omega(\mathbf{R}) = i \sum_{\psi_n \neq \psi} \langle \partial_x \psi | \psi_n \rangle \langle \psi_n | \partial_y \psi \rangle + h.c. \quad (3.30)$$

Using the identity,

$$\langle \psi_n | \partial_{\alpha} \psi \rangle = \frac{\langle \psi_n | \partial_{\alpha} H | \psi \rangle}{\epsilon - \epsilon_n}, \text{ for } \psi \neq \psi_n, \quad (3.31)$$

we can re-express the Berry curvature in the form

$$\Omega(\mathbf{R}) = i \sum_{\psi_n \neq \psi} \frac{\langle \psi | \partial_x H | \psi_n \rangle \langle \psi_n | \partial_y H | \psi \rangle}{(\epsilon - \epsilon_n)^2} + h.c. \quad (3.32)$$

2. Note that the definition provided for the Berry curvature (eqn. 3.29) is only well-defined across $\mathcal{R}$ when the state remains fully gapped. That is, no other states ever have the same energy as $|\psi(\mathbf{R})\rangle$. When another state energy crosses that of $|\psi\rangle$, the individual states become ill-defined and so it becomes challenging to write down a natural derivative for the states. With this in mind, let us consider re-expressing the Berry curvature in terms of the single state projector $\hat{\psi} = |\psi\rangle \langle \psi|$,

$$i \langle \partial_x \psi | \partial_y \psi \rangle + h.c. = i \text{Tr} \left[\hat{\psi} \partial_x \hat{\psi} \partial_y \hat{\psi} \right] + h.c. \quad (3.33)$$

For a single state, this is identical to the regular Berry curvature, however the utility of this expression becomes evident when we construct it for a projector onto a set of states $P = \sum_{i \in \text{band}} |\psi_i\rangle\langle\psi_i|$. Since ∂P is still well-defined when the states in P are degenerate, we can write an expression for the combined berry curvature of a set of states that can handle degeneracy (provided no states outside P become degenerate with P).

$$\Omega_P = i\text{Tr}[P\partial_x P\partial_y P] + h.c. \quad (3.34)$$

This calculation has one important caveat. It is only possible to apply Stokes' theorem when $\mathbf{A}$ is smooth over the region $\mathcal{R}$. This means that we need to have already picked a smooth gauge in the region we are integrating over to have a guarantee that eqns. 3.25 and 3.25 hold. If we transform to a non-smooth gauge then it is possible that the Berry phase will change by a multiple of 2π , however the Berry curvature is truly gauge invariant. This can be seen from eqn. 3.32, where the Berry curvature is defined without the differentiation of states, so is invariant under even a non-smooth gauge transformation. Thus in general we only have the guarantee that

$$e^{-i\gamma(\mathcal{C})} = e^{\int_S d^2 R \Omega(\mathbf{R})}. \quad (3.35)$$

Chern Number

As in §3.1.1, we make the assertion that the $\mathbf{R}$ -space lives on the torus, a closed manifold. The Chern number can then be defined as the integral of the Berry curvature across the whole manifold,

$$\mathcal{Q} = \frac{1}{2\pi} \int d^2 R \Omega. \quad (3.36)$$

As in the discrete case, the Chern number here is restricted to only take integer values. The argument follows the same lines as previously, where we can use our modified form of Stokes theorem (eqn. 3.35) to show that the Chern number is equivalent to the Berry phase around a loop enclosing no area, implying that

$$e^{\int d^2 R \Omega(\mathbf{R})} = 1, \implies \mathcal{Q} \in \mathbb{Z} \quad (3.37)$$

3.1.3 An Example – The Two-Level System

Let us illustrate these concepts with the example of a two-state system. Our first step is to parametrise the possible Hamiltonians in such a system using a vector $\mathbf{d}$. We will then solve this Hamiltonian for the positive and negative energy eigenstates. This allows us to look at how the eigenstates change as we smoothly deform the Hamiltonian, exploring the parameter space of $\mathbf{d}$. By examining how the eigenstates of $\hat{H}$ depend on the parametrisation, we can calculate the Berry curvature of the positive and negative energy states. We then connect this with our understanding of the Bloch states in a periodic quantum system. We discuss how the bulk

Hamiltonian of a periodic system corresponds to the embedding of a closed surface in $\mathbf{d}$ -space and describe how to calculate the Chern number of the bands in such a system.

The Hilbert space of a two-level system is $\mathbb{C}^2$. This means that any Hamiltonian we construct must be a 2×2 Hermitian matrix. We will parametrise this in terms of a Pauli vector, $\mathbf{d}$,

$$H = \mathbf{d} \cdot \hat{\sigma} = d_i \sigma^i. \quad (3.38)$$

where the σ^i are the Pauli matrices, which form a complete basis for 2×2 traceless hermitian matrices. Of course, we could include a multiple of the identity to span the full space of Hermitian matrices, however this would have no effect on the topology so we shall omit it.

The set of traceless Hamiltonians is thus a manifold in itself, $\{H\} \cong \mathbb{R}^3$, since $\mathbf{d} \in \mathbb{R}^3$. The first thing we will do is to express the vector $\mathbf{d}$ in standard spherical polar coordinates, giving us a Hamiltonian that looks like

$$H = |\mathbf{d}| \begin{pmatrix} \cos(\theta) & e^{-i\phi} \sin \theta \\ e^{i\phi} \sin \theta & -\cos(\theta) \end{pmatrix} \quad (3.39)$$

This matrix has $H^2 = |\mathbf{d}|^2 \mathbb{1}$, so its eigenvalues are $\pm |\mathbf{d}|$. One can show that its eigenvectors are

$$|+\mathbf{d}\rangle = \begin{pmatrix} e^{-i\phi/2} \cos(\theta/2) \\ e^{i\phi/2} \sin(\theta/2) \end{pmatrix}, \quad |-\mathbf{d}\rangle = \begin{pmatrix} -e^{-i\phi/2} \sin(\theta/2) \\ e^{i\phi/2} \cos(\theta/2) \end{pmatrix}. \quad (3.40)$$

We have thus arrived at a set of states, $|\pm\mathbf{d}\rangle$ that depend on some parameter $\mathbf{d}$. This means that we can apply the techniques of the last section. From eqns. 3.21 and 3.26 we know that the Berry connection and Berry curvature of the $|\pm\rangle$ states are given by

$$\mathbf{A}^\pm(\mathbf{d}) = i \langle \pm\mathbf{d} | \nabla_{\mathbf{d}} | \pm\mathbf{d} \rangle \quad (3.41)$$

$$\mathbf{\Omega}^\pm(\mathbf{d}) = \nabla_{\mathbf{d}} \wedge \mathbf{A}^\pm(\mathbf{d}). \quad (3.42)$$

Using this equation it is possible to directly calculate the Berry curvature, arriving at the following expression:

$$\mathbf{\Omega}^\pm = \pm \frac{\hat{n}_{\mathbf{d}}}{2\mathbf{d}^2} \quad (3.43)$$

where $\hat{n}_{\mathbf{d}}$ is a unit vector pointing in the direction of $\mathbf{d}$.

The Berry curvature is equivalent to a monopole field radiating out from the origin in $\mathbf{d}$ -space, in analogy with electrostatics. Thus, the Berry phase of the $|+\mathbf{d}\rangle$ state around some path in $\mathbf{d}$ -space is given by

$$\gamma^+(C) = \oint_C \mathbf{A}^+(\mathbf{d}) \cdot d\mathbf{d}. \quad (3.44)$$

Re-expressing this in terms of the Berry curvature gives us

$$\gamma^+(C) = \int_{\mathcal{R}} \mathbf{\Omega}^+(\mathbf{d}) \cdot d\mathbf{S}, \quad (3.45)$$

where the integral is over a region $\mathcal{R}$ that has the curve $\mathcal{C}$ as its boundary. Since the Berry curvature is just a monopole field with charge $\frac{1}{2}$, this integral ends up evaluating half the solid angle subtended by the curve $\mathcal{C}$.

Chern Number

Now we can connect this with our understanding of Bloch states. For a two-dimensional translationally symmetric system, the Hamiltonian can be diagonalised according to

$$H = \sum_{\mathbf{k}} |\mathbf{k}\rangle \langle \mathbf{k}| \otimes H(\mathbf{k}). \quad (3.46)$$

If the unit cell contains two sites, then $H(\mathbf{k})$ is a 2×2 matrix that depends on $\mathbf{k}$. We can write it in terms of a $\mathbf{d}$ vector that depends on $\mathbf{k}$.

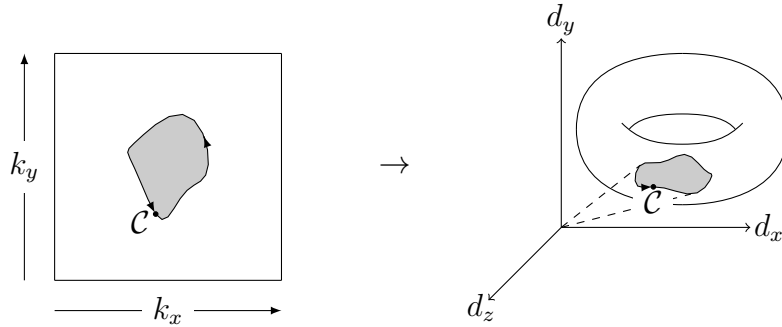
$$H(\mathbf{k}) = \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}. \quad (3.47)$$

$\mathbf{k}$ lives on the surface of a torus, since k_x and k_y are periodic. Furthermore, $\mathbf{d}$ (and by extension H) live in the space $\mathbb{R}^3$. Thus, the mapping $\mathbf{d}$ is

$$\mathbf{d} : S^1 \times S^1 \rightarrow \mathbb{R}^3. \quad (3.48)$$

The map is embedding a two-dimensional surface (the torus) into three-dimensional flat space ($\mathbb{R}^3$). There are no limits on how you can do this embedding, the torus can take any shape provided the embedding respects the topology of the Brillouin zone.

To calculate the Berry phase around a loop in $\mathbf{k}$ -space, all we have to do is look at the corresponding loop in $\mathbf{d}$ -space. Then we integrate the Berry connection over some surface that has this loop as its boundary. The Berry connection actually looks like a radiating monopole so we can simplify further, knowing that we just need to find the solid angle subtended by the curve around the origin and divide it by two.



What about the Chern number? To calculate this all we need to do is integrate the Berry curvature across the whole surface of the torus in $\mathbf{d}$ -space. If the embedding of the torus does not contain the origin, then the value of this integral is necessarily zero.

Now suppose the torus is embedded such that it contains the origin, then we will have a surface that catches all of the Berry flux on its inside and the solid angle subtended is 4π ! This means that the Chern number will be $+1$. Furthermore, you could imagine wrapping the torus around the origin twice, so that it effectively contains two origins, now the Chern number will be $+2$. You could turn it inside out and then wrap it around the origin too, that would give $\mathcal{Q} = -1$ and so on...

3.1.4 Final Notes

this section needs a bit more formal equation-y stuff

We close this section with an explanation of the consequences of Chern number in topological insulators.

Smoothly Deforming the Hamiltonian

Now suppose that we have a pair of quantum systems that live in the same Hilbert space, described by $\hat{H}_1$ and $\hat{H}_2$. Suppose both of these Hamiltonians represent insulating materials. This means that the bands in these materials are separated across all of the Brillouin zone by a band gap. We might want to ask the question: *Is it possible to smoothly deform $\hat{H}_1$ into $\hat{H}_2$ without ever crossing a point where the bands touch and the material becomes conductive?*

The answer to this is that it is only possible if the bands of $\hat{H}_1$ and $\hat{H}_2$ have the same Chern numbers. This is because the Chern number is always restricted to have integer value, it cannot continuously change as we smoothly deform the Hamiltonian. In order to change the Chern number of a band we must somewhere cross a point where it is impossible to define a Chern number at all, allowing it to jump discontinuously when we leave that point. This is precisely what happens when two bands touch, the description of a single band becomes ill-defined and our calculation for Chern number stops working.

We can illustrate this with the example in §3.1.3. Smoothly deforming our Hamiltonian amounts to changing the shape of the torus embedded in $\mathbf{d}$ -space. If any point on the surface of the torus touches the origin, we have a pair of zero-energy states appearing in our system, this means the bands touch and the system is conductive. If the band has a Chern number of one, this means that the torus contains the origin. If we wanted to smoothly deform this to have zero Chern number, we would need to move the torus such that the origin was now outside. At some point in this deformation the torus surface will have to touch the origin, creating a conducting Hamiltonian at that point.

Thus we can classify all Hamiltonians according to the Chern number of their bands. It is always impossible to smoothly deform a Hamiltonian into one with a different Chern number while remaining insulating. Similarly, it is always possible to smoothly deform two Hamiltonians into one another without closing a gap, provided that their bands have identical Chern numbers [3].

This allows us to make a rough justification for why conducting edge states must appear wherever we have a material with nonzero Chern number. Suppose we have a large material where the Hamiltonian is allowed to vary slowly over space, defined by $\hat{H}(\mathbf{r})$. Now let's say that in some region this Hamiltonian looks locally like it has non-zero Chern number, $\hat{H}_{\text{topological}}$, but in a different, distant region it looks locally like it has zero Chern number, $\hat{H}_{\text{trivial}}$. We expect that between these regions the Hamiltonian must be gradually transforming between $\hat{H}_{\text{topological}}$ and $\hat{H}_{\text{trivial}}$. As we have described, it is impossible for our Hamiltonian to make this transformation without creating a conducting surface somewhere between these two regions – edge states must appear! Loosely speaking, we can view the vacuum as a zero Chern number material, which gives us an intuitive picture for why we might expect materials with non-zero Chern number to always have edge states appearing on their surface, where the Hamiltonian has to transition to the vacuum.

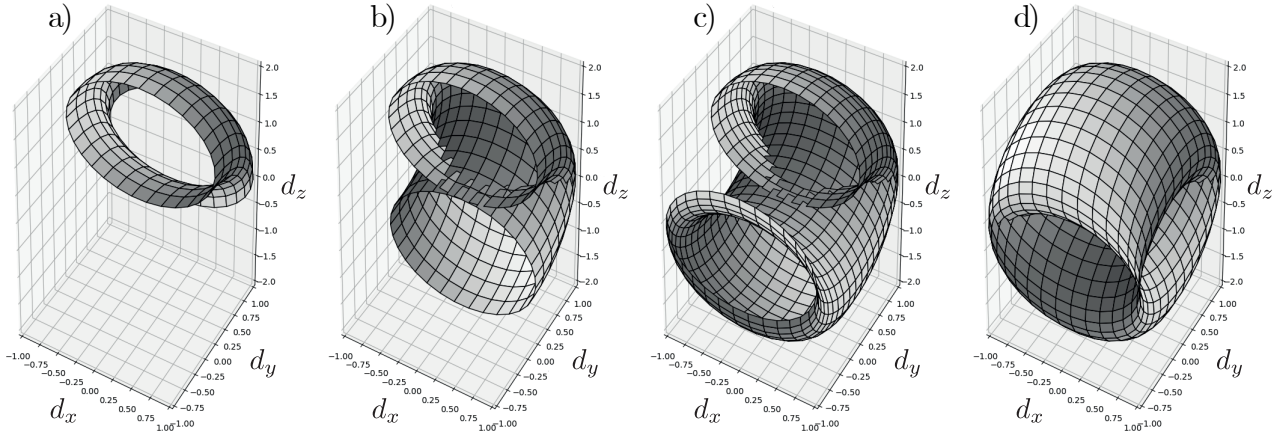


Figure 3.2: The embedding of the Brillouin zone in $\mathbf{d}$ -space for the QWZ model. This is shown in four progressive cutaways to ensure the internal structure can be seen. In all four images k_x spans the interval $[0, 2\pi)$, however in (a) k_y spans $[0, \frac{\pi}{2})$, in (b) k_y spans $[0, \pi)$, in (c) k_y spans $[0, \frac{3\pi}{2})$, and in (d) k_y spans $[0, 2\pi)$.

Chern Number and Globally Smooth Gauges

If you thought carefully about the gauge that we chose in equation 3.40 you might notice that it has a problem, There is actually a discontinuity at $\phi = \pi$ where $e^{i\phi}$ suddenly jumps as you cross over to $\phi = -\pi$. This is no oversight, it turns out that it is actually completely impossible to define a globally smooth set of states $|\psi(\mathbf{d})\rangle$ for any band that has nonzero Chern number!

This is because if $|\psi(\mathbf{d})\rangle$ were continuous everywhere, then the Berry connection would also have to be continuous everywhere. This would mean that we could invoke the exact Stokes' theorem when calculating the Chern number. Since the Berry curvature is integrated is over the whole of $\mathbf{k}$ -space it must have no boundary, $\mathcal{Q}$ must be equal to zero.

In this sense, the Chern number can be regarded as something like a measure of the amount of discontinuity that *must* be present regardless of the gauge you choose for the space.

3.2 The Qi-Wu-Zhang Model

Let us introduce the Qi-Wu-Zhang (QWZ) model as a paradigmatic example of a two-dimensional topological insulator with nearest-neighbour interactions [4, 5].

add some history on the model here

We work on a square lattice of unit cells indexed by a two dimensional lattice vector $\mathbf{m}$, where each cell contains two sites. The QWZ model is defined by three terms, a hopping between unit cells in the x -direction, a hopping in the y -direction, and a staggered on-site potential whose

strength is determined by a parameter u .

$$\begin{aligned}\hat{H} = & \sum_{\mathbf{m}} |\mathbf{m} + \mathbf{1}_x\rangle \langle \mathbf{m}| \otimes \frac{\sigma_z + i\sigma_x}{2} + h.c. \\ & + \sum_{\mathbf{m}} |\mathbf{m} + \mathbf{1}_y\rangle \langle \mathbf{m}| \otimes \frac{\sigma_z + i\sigma_y}{2} + h.c. \\ & + \sum_{\mathbf{m}} |\mathbf{m}\rangle \langle \mathbf{m}| \otimes u \sigma_z,\end{aligned}\tag{3.49}$$

where $\mathbf{1}_x$ and $\mathbf{1}_y$ represent the unit displacement in their respective directions,

$$\mathbf{1}_x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{1}_y = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.\tag{3.50}$$

Using Bloch's theorem, we can find the bulk Hamiltonian,

$$\hat{H}(\mathbf{k}) = \begin{pmatrix} \sin k_x \\ \sin k_y \\ u + \cos k_x + \cos k_y \end{pmatrix} \cdot \boldsymbol{\sigma}.\tag{3.51}$$

Here, k_x and k_y determine our position in the Brillouin zone. As explained in §3.1.3, this represents the embedding of a torus in $\mathbf{d}$ -space, shown in fig. 3.2.

The u -parameter has the effect of translating the torus by a fixed amount in the z -direction. From this we can see that the QWZ model only has non-zero Chern number for values of u between -2 and 2. Translating the torus by any more will move the origin completely outside. Thus the Chern number is

$$Q = \begin{cases} -1 & : -2 < u < 0, \\ 1 & : 0 < u < 2, \\ 0 & : \text{otherwise}, \end{cases}\tag{3.52}$$

not including the points at $u = -2, 0$ and 2 , at which the Chern number is undefined because the band gap closes and the material becomes conducting. Some example band structures for varying values of u are shown in fig. 3.3.

3.2.1 Strip Geometry and Edge Modes

According to the bulk-boundary correspondence, whenever the Chern number $\mathcal{C}$ is nonzero, the QWZ model must have gapless edge modes appearing. To provide an example, here we construct the model on strip geometry and provide an analytical expression for the edge modes.

We shall require that the system is periodic and uniform in the y -direction, but is allowed to be non-uniform and have open boundaries in the x -direction. We allow u to vary in the x -direction, so add a subscript to indicate position-dependence, $u \rightarrow u_{m_x}$. We can apply Bloch's theorem in the y -direction only, and our eigenstates take the following form,

$$|\psi\rangle = |k_y\rangle \otimes |v_{k_y}\rangle,\tag{3.53}$$

where $|k_y\rangle$ is a plane wave in the y direction and $|v_{k_y}\rangle$ is a $2 \times L_x$ element vector. The v -states

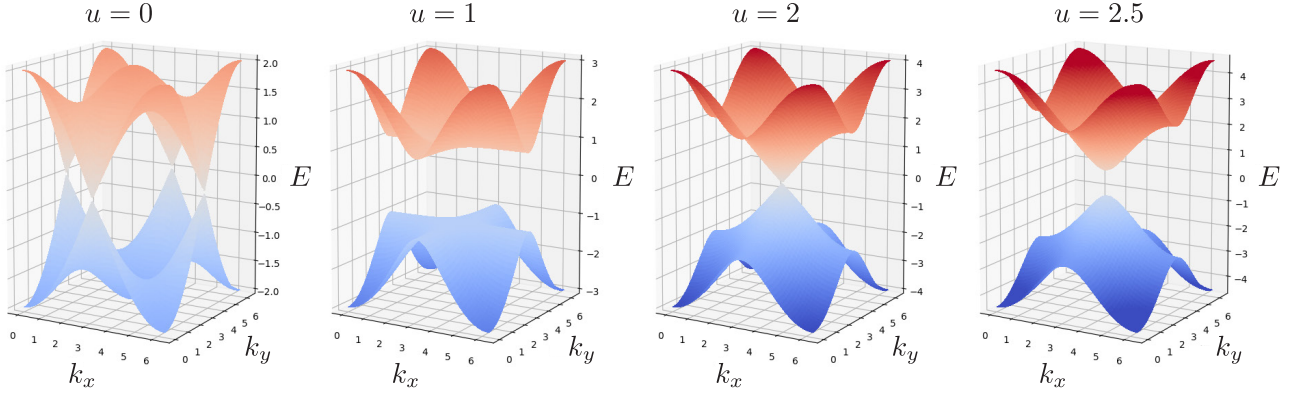


Figure 3.3: Four examples of the band structure of the QWZ model are shown, for varying u values between 0 and 2.5. At $u = 0$ and 2, the bands touch and Chern number is undefined. At $u = 1$, $\mathcal{C} = +1$, and at $u = 2.5$, $\mathcal{C} = 0$.

are solutions of the semi-bulk Hamiltonian

$$\begin{aligned} \hat{H}(k_y) = & \sum_{m_x} |m_x + 1\rangle \langle m_x| \otimes \frac{\sigma_z + i\sigma_x}{2} + h.c. \\ & + \sum_{m_x} |m_x\rangle \langle m_x| \otimes [(u_{m_x} + \cos k_y) \sigma_z + \sin k_y \sigma_y]. \end{aligned} \quad (3.54)$$

We have effectively reduced our system from a lattice of sites in x and y , to a set of chain models in x , that are each labelled by a value of k_y . The band structure for this system in both periodic boundaries and strip geometry is shown in fig. 3.4. Note that edge states are only present for certain values of k_y , which can be seen in figure 3.4, where the edge states peel away from the bulk in a region around $k_y = \pi$.

The strip Hamiltonian can be solved by proposing modes that decay exponentially away from the boundary. Thus, a left-edge state has the form

$$|\psi_{k_y}^{\text{left}}\rangle = \sqrt{1 - \xi_l^2} \sum_{m=1}^{L_x} \xi_l^{m-1} |m\rangle \otimes \begin{pmatrix} 1 \\ -i \end{pmatrix} \quad (3.55)$$

where

$$\xi_l = -u - \cos k_y. \quad (3.56)$$

This state has energy

$$E_l = -\sin k_y. \quad (3.57)$$

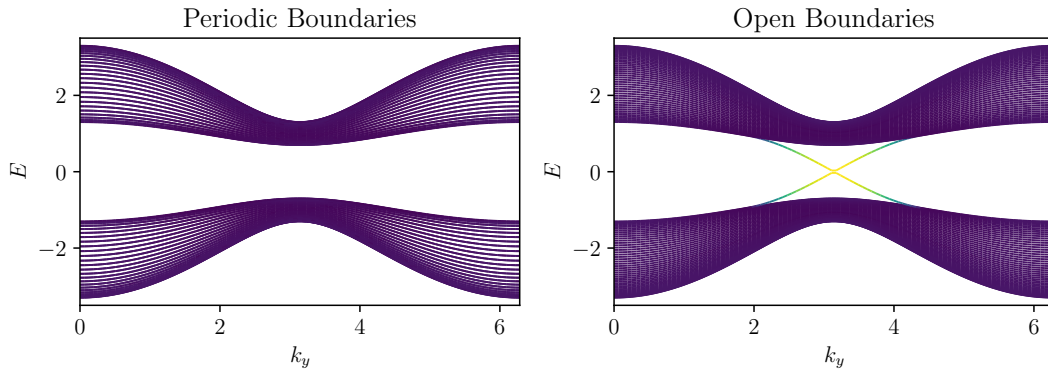


Figure 3.4: An example band structure for a QWZ model with $u = 1.3$ and $L_x = 30$. States are coloured according to their inverse participation ratio, indicating their relative degree of localisation. **(Left)** The example for periodic boundary conditions, showing two well-separated bands. **(Right)** The case for open boundary conditions. Two edge states are shown leaving the bulk and crossing the gap, with the descending state being a left edge state, and the ascending state being a right edge state.

Similarly, the right edge state is determined by

$$|\psi_{k_y}^{\text{right}}\rangle = \sqrt{1 - \xi_r^2} \sum_{m=1}^{L_x} \xi_r^{L_x-m} |m\rangle \otimes \begin{pmatrix} 1 \\ i \end{pmatrix} \quad (3.58)$$

$$\xi_r = -u - \cos k \quad (3.59)$$

$$E_r = \sin k. \quad (3.60)$$

These edge states are only valid solutions for k_y in the region

$$-1 < u + \cos k_y < 1. \quad (3.61)$$

These limits can be seen in fig. 3.4, where the edge states peel away from the bands to cross the energy gap. These expressions are only strictly valid in the limit of a large system. since we have made the assumption the edge states can be calculated by taking into consideration only one edge and ignoring the effect of the other. The edge states decay exponentially through the system, so we are relying on the assumption that the edge state has effectively decayed to nothing before reaching the other side. When this condition is relaxed the edge states are able to couple to one another. This has a two effects, the first being that at the crossing of edge states, our energy eigenvalues become positive and negative superposition states of both edge states. The other is that the left state coupling to the right state shifts their energy levels producing an avoided crossing such that no states cross $E = 0$.

3.2.2 The Amorphous QWZ Model

We now look at extending the QWZ model beyond a grid to arbitrary amorphous lattices. This is done by defining a smooth continuation of the QWZ model to allow for edges that can point in an arbitrary direction. Given an arbitrary lattice, we first require that every vertex hosts two states, and the on-site energy term containing $u\sigma_z$ is identical as in eqn 3.49. To construct an

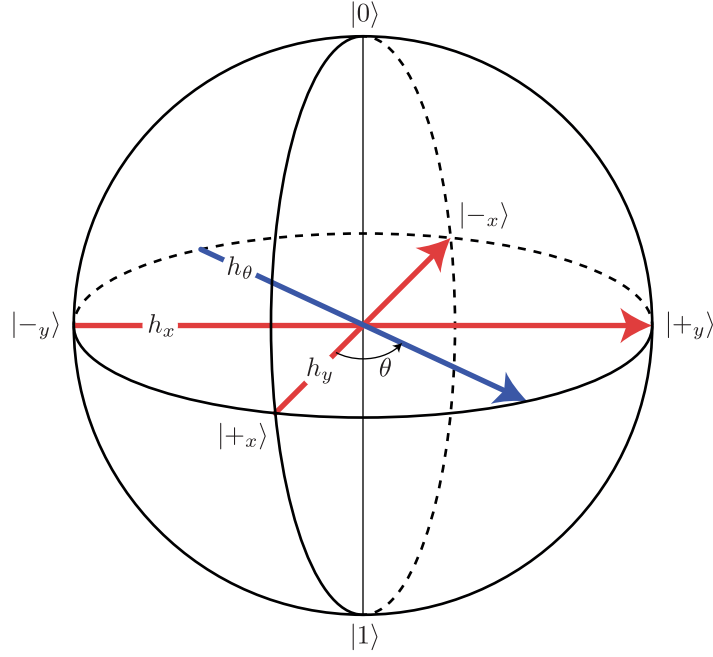


Figure 3.5: The Bloch sphere, showing the positions of the eigenstates of the three Pauli matrices. The action of the hopping matrices is shown in red, where the h_x and h_y operators act as creation operators on the y and x bases respectively. The action of our arbitrary angle hopping operator is shown in blue, acting as a creation operator between a pair of orthogonal states that lie on the x - y plane.

amorphous equivalent of the hopping terms, let us first look at the hopping in the QWZ model more carefully. We have two hopping matrices, one that hops a state in the x -direction, h_x , and one that hops in the y -direction, h_y .

$$h_x = \frac{\sigma_z + i\sigma_x}{2} = \frac{1}{2} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \quad (3.62)$$

$$h_y = \frac{\sigma_z + i\sigma_y}{2} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}. \quad (3.63)$$

These operators can be viewed as creation operators on different bases, h_x acts on the basis of σ_y , and h_y acts on the basis of σ_x . Viewed in the Bloch sphere, these operators hop the state across the sphere in the x or y direction.

In our amorphous lattice, hopping operators can act across an edge connecting two sites that points in an arbitrary direction. Thus, rather than trying to assign edges with h_x or h_y operators, let us define an arbitrary-angle hopping matrix h_θ , that hops in a direction θ over the x - y plane of the Bloch sphere,

$$h_\theta = |+\theta\rangle \langle -\theta| \quad (3.64)$$

$$= \frac{1}{2} \begin{pmatrix} 1 & -e^{-i\theta} \\ e^{i\theta} & -1 \end{pmatrix}. \quad (3.65)$$

Thus, the Hamiltonian we use for the amorphous system is given by

$$\begin{aligned}\hat{H} = & \sum_{i,j} |\mathbf{r}_i\rangle \langle \mathbf{r}_j| \otimes h_{\theta_{i,j}} + h.c. \\ & + \sum_i |\mathbf{r}_i\rangle \langle \mathbf{r}_i| \otimes u_i \sigma^z,\end{aligned}\tag{3.66}$$

where $\theta_{i,j}$ is the angle between the edge connecting $\mathbf{r}_i$ and $\mathbf{r}_j$ and the x axis.

As before, the parameter u controls whether the system is in a topological or trivial state. The exact dependence of the system on u depends strongly on the lattice geometry, however there is generally a region around which the system has nonzero topological character. In the absence of a momentum-space description, the system's topological character can be seen indirectly by comparing the spectrum in open and periodic boundary conditions. Generally one expects the presence of a gap in periodic boundaries that becomes populated with edge modes in open boundaries. Two examples are shown in fig. 3.6.

3.2.3 Topology of the Amorphous QWZ Model

OPTIONAL – you could add a section here on the low- k theory of the amorphous QWZ model if you wanted - would allow you to make some predictions for the Chern number as you can find the berry phase change around the low k band touching. Maybe its a bit overkill - the thesis looks like its gonna be kinda long anyway.

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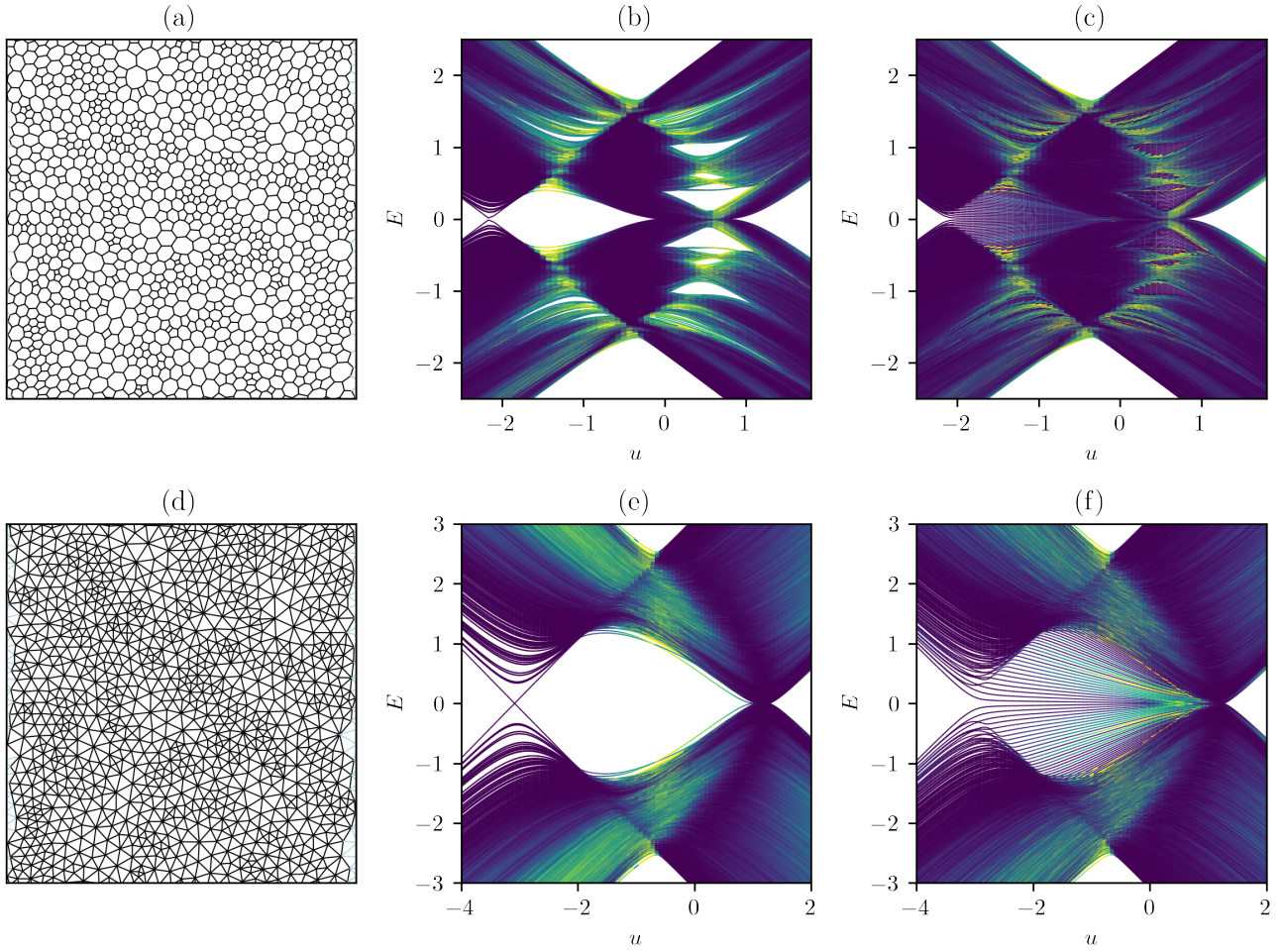


Figure 3.6: Plots showing the spectra of the amorphous QWZ model for two different example lattices. States are coloured according to their inverse participation ratio. **(a)** A trivalent lattice generated from a Voronoi decomposition of a set of random points, followed by five iterations of Lloyd's algorithm to regularise bond lengths. **(b)** The spectrum of the QWZ model on the voronised lattice as a function of u . Several gaps are present at different values of u and E . **(c)** The spectrum of the same model as (b), but in open boundary conditions. Here we can see that all gaps for $u \in [-1.5, 1]$ are now filled with edge modes, indicating that the system is in a topological phase. **(d)** An example of a lattice forming a Delaunay triangulation, constructed as the dual of the lattice in (a). **(e)** The spectrum of the QWZ model on the Delaunay lattice as a function of u showing one single large gapped region. **(f)** The spectrum of the same model as (e), but in open boundary conditions. Here we can see that the gaps is now filled with edge modes, again indicating that here the system is in a topological phase.

Chapter 4

Local Markers

The concepts outlined in Chapter 3 – Berry phase, curvature and Chern number – were first proposed in the 1980s to explain the newly discovered quantum Hall effect [1, 2], where they provided the mathematical underpinning for the TKNN invariant that defines the Hall conductance [3, 4]. In the decades since, the Chern number has been one of the central tools for classifying condensed matter systems outside of the Landau symmetry-breaking framework [5–8]. They have found use in describing ocean waves [9] coupled mechanical systems [10–12], as well as providing the foundations for the modern theory of polarisation [13–15].

One of the defining characteristics of quantum Hall physics is the robustness of the edge states and Hall conductance to impurities in the sample, where certain amount of disorder is even *required* to stabilise the quantised conductivity [16, 17]. Despite this, our definition of the Chern number cannot be calculated in a disordered material. In order to calculate the Berry curvature – and thus the Chern number – we must work in momentum space. If a material lacks translational symmetry, it is not possible to apply Bloch’s theorem, crystal momentum is not a good quantum number and we simply have no concept of momentum space. Disorder-resistant methods exist [18, 19], however they compute a global Chern number, and cannot capture the local properties of materials with compound structure. This tension, that our tools for describing disorder-resistant physics are themselves undermined by disorder, suggests that our mathematical framework is incomplete.

One potential solution to this problem is the development of real space methods for calculating the Chern number. These attempt to translate the Chern number into a mathematical language that is resolved in real space, ensuring it is well-defined when the system has no translational symmetry. There are a litany of proposals, such as Kitaev’s Chern marker [12, 20], the Bott Index [21–24], the Chern marker [25, 26], the localizer index [27] and most recently structural spillage [28]. In all of these cases, the aim is to re-express the definition of Chern number in terms of a set of real-space operators which are local, and thus do not rely on the notion of momentum space to make sense. Such a quantity should be equivalent to the Chern number when calculated in a translationally symmetric context; however would still be possible in cases where these conditions have been relaxed, and the notion of k -space is gone. These quantities afford the researcher tools with which to comment on the topological characteristics of materials that had previously been out of reach, such as amorphous crystals [29], quasi-periodic lattices [30], systems out of equilibrium [22, 31–33] and even strongly interacting systems [34].

In this section we start by introducing three such quantities, the Kitaev Chern number, the Bott index and the Chern marker, giving their definitions as well as explaining how they are related.

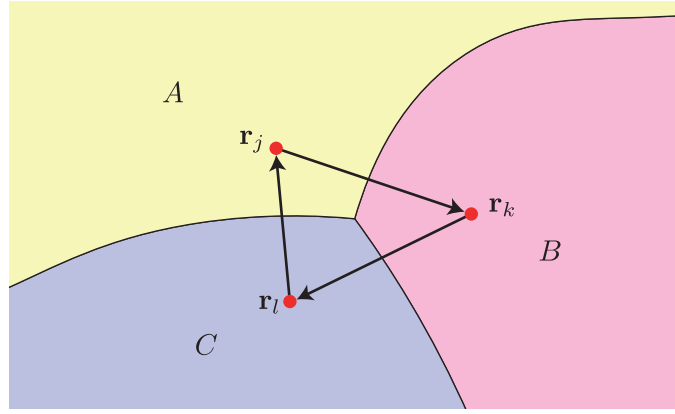


Figure 4.1: Diagram showing the partition of the plane into three regions labelled with A , B and C . An example of h_{jkl} is depicted with red dots, where arrows indicate the positive direction.

add a bit more summary of wot u writing

4.1 Kitaev's Local Marker

The majority of accounts of local markers tend to focus on the development of the Bott index and the Chern marker. However, the first construction of a real-space Chern number can, along with the rest of modern physics, be found in the appendices of a seminal work by Kitaev [20]. Thus, we shall start our discussion with an explanation of Kitaev's local marker, the discussion will largely follow the source material, however we provide some detail at various points where it is brushed passed there.

As always, our starting point is a Hamiltonian defined on an arbitrary lattice with sites at positions $\mathbf{r}_j$, however in this case we shall work in the case of a system with infinite boundary conditions. Here, let us define the projector onto a band $P = \sum_{\text{band}} |\psi\rangle\langle\psi|$, along with a shorthand for off-diagonal correlations given by

$$P_{jk} = \langle \mathbf{r}_j | P | \mathbf{r}_k \rangle. \quad (4.1)$$

In general, if this is onto a gapped set of states, then the projector will have exponentially localised correlations [35],

$$|P_{jk}| \leq c |\mathbf{r}_j - \mathbf{r}_k|^{-\alpha} \quad (4.2)$$

maybe add an argument for why this has to be the case – see the Paley-Wiener theorem

Following Kitaev, let us define a 2-current,

$$h_{jkl} = 12\pi i (P_{jk}P_{kl}P_{lj} - P_{jl}P_{lk}P_{kj}). \quad (4.3)$$

We divide our system into three regions in real space, A , B and C , and define the real-space Chern number,

$$\mathcal{Q}_{\text{Kit}}(P) = \sum_{j \in A} \sum_{k \in B} \sum_{l \in C} h_{jkl}, \quad (4.4)$$

which we can re-express in terms of a set of projectors onto the three regions,

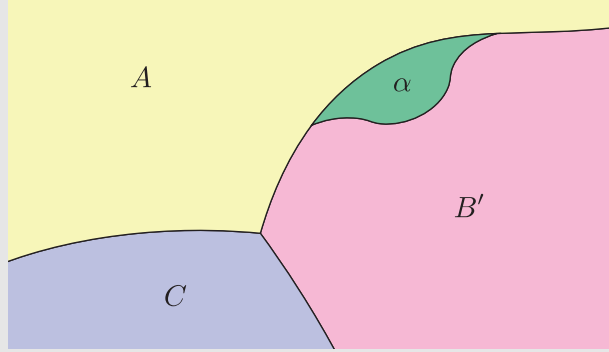
$$\mathcal{Q}_{\text{Kit}}(P) = 12\pi i \text{Tr}(PAPBPC - PCPBPA). \quad (4.5)$$

This quantity remains invariant when we relabel any finite area from one region to another.

The Invariance of $\mathcal{Q}_{\text{Kit}}$ to changes in A , B and C — In order to see why this is the case, let us consider picking up some finite chunk of the region B , which we label with α and moving it over into region A . Thus, let us make the redefinition

$$A' = A + \alpha \quad (4.6)$$

$$B' = B - \alpha \quad (4.7)$$



Thus, the matrix term in the Kitaev Chern number becomes

$$PAPBPC - h.c. = PA'PB'PC + PAP\alpha PC - P\alpha PB'PC - h.c. \quad (4.8)$$

Now let us use the identity $B' = \mathbb{1} - \alpha - A - C$ to re-express this in the form

$$\begin{aligned} PAPBPC - h.c. = & PA'PB'PC + PAP\alpha PC + P\alpha PAPC \\ & - P\alpha PC + P\alpha P\alpha PC + P\alpha PCPC - h.c. \end{aligned} \quad (4.9)$$

where we have used the identity $P^2 = P$ to obtain this line. All the extra terms on the RHS vanish when the trace is taken. Note that generally one cannot use the cyclic property of the trace here, since our operators are infinite matrices, so there is no guarantee that the rearranged infinite sum converges to the same value. However, since α represents a finite region, the sum vanishes far from α and so we are free to cycle any trace containing α . Thus we are left with

$$\text{Tr}(PAPBPC - h.c.) = \text{Tr}(PA'PB'PC - h.c.). \quad (4.10)$$

In fact, the extra finite region α need not be a projector, since the above proof holds regardless of the properties of α provided that it has finite support. Thus we can go even

further and see that the value of $\mathcal{Q}_{\text{Kit}}$ is invariant under the addition of any finite matrix to one region, provided we subtract it from another region as well.

Let us re-express $\mathcal{Q}_{\text{Kit}}$ to a form that will be useful in the following discussion. Since the expression is invariant under cycling of the order of our three regions, we may write it as

$$\mathcal{Q}_{\text{Kit}} = 4\pi i \text{Tr}(PAPBPC + PBPCPA + PCPAPB) \quad (4.11)$$

$$- PAPCPB - PCPBPA - PBPAPC), \quad (4.12)$$

note that we are not allowed to cycle the terms in the trace here. Inserting the identity

$$C = \mathbb{1} - A - B, \quad (4.13)$$

we arrive at the following expression,

$$\mathcal{Q}_{\text{Kit}}(P) = 4\pi i \text{Tr}[PAP, PBP] \quad (4.14)$$

This is a trace of a commutator, which will always vanish if one of the operators in the commutator has finite support. This supports our assertion that $\mathcal{Q}_{\text{Kit}}$ is invariant under addition of any finite matrix to $A \rightarrow A + \alpha$, since $\text{Tr}[P\alpha P, PBP]$ vanishes. Of course we must also transform either $B \rightarrow B - \alpha$ or $C \rightarrow C - \alpha$ for the identity 4.13 to hold.

This invariance to the form of A, B and C affords us with an astounding amount of freedom in how we choose to define this Chern number. Not only can we select any three regions with any shape (provided the overall topology of the three regions far from the triple contact point remains the same). We can even blur the boundaries between the projectors or allow them to overlap and nothing will change.

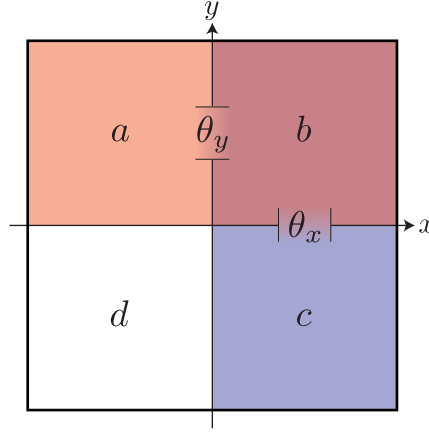
4.1.1 Relation to the Chern Number

So far we have constructed a quantity, $\mathcal{Q}_{\text{Kit}}$ and shown that there is a remarkable amount of freedom in how we can build it without changing the value. In order to connect this to band topology, we will need to show that when evaluated in a translationally symmetric context, $\mathcal{Q}_{\text{Kit}}$ corresponds exactly to the Chern number. Let us start by re-expressing $\mathcal{Q}_{\text{Kit}}$ again in a new form. We define a set of step function projectors ϑ_x, ϑ_y that satisfy

$$\vartheta_\alpha |\mathbf{r}\rangle = \begin{cases} |\mathbf{r}\rangle & \text{for } r_\alpha \geq 0, \\ 0 & \text{for } r_\alpha < 0. \end{cases} \quad (4.15)$$

Let us now consider the operator

$$2\pi i \text{Tr}[P\vartheta_x P, P\vartheta_y P]. \quad (4.16)$$



By decomposing the step function into projectors onto the four quarters of the plane, we arrive at

$$2\pi i \text{Tr}[P\vartheta_x P, P\vartheta_y P] = 2\pi i \text{Tr}([PaP, PbP] + [PbP, PbP] + [PaP, PcP] + [PbP, PcP]) \quad (4.17)$$

Clearly the second term on the RHS vanishes. The third term also vanishes, which can be seen by its invariance under a mirror symmetry across the line $y + x = 0$ (a mirror reflection will always change the sign of h_{jkl}). Finally, we see that the first and last terms are both equal to $\mathcal{Q}_{\text{Kit}}/2$, thus

$$\mathcal{Q}_{\text{Kit}} = 2\pi i \text{Tr}[P\vartheta_x P, P\vartheta_y P]. \quad (4.18)$$

In order to make contact with momentum space, we shall work in an infinite translational system. Clearly, in this context the position of the step functions will have no effect on the value of $\mathcal{Q}_{\text{Kit}}$. Thus, let us define a set of translated step functions

$$\vartheta_\alpha(\mathbf{R}) |\mathbf{r}\rangle = \begin{cases} |\mathbf{r}\rangle & \text{for } r_\alpha \geq R_\alpha, \\ 0 & \text{for } r_\alpha < R_\alpha. \end{cases} \quad (4.19)$$

Obviously, the value of $\mathcal{Q}_{\text{Kit}}(\mathbf{R}) = 2\pi i \text{Tr}[P\vartheta_x(\mathbf{R})P, P\vartheta_y(\mathbf{R})P]$ is constant. Thus let us choose some square region of area L^2 the plane and take the average of $\mathcal{Q}_{\text{Kit}}(\mathbf{R})$ over this region,

$$\mathcal{Q}_{\text{Kit}} = \frac{1}{L^2} \int_{-L/2}^{L/2} dR_x dR_y \mathcal{Q}_{\text{Kit}}(\mathbf{R}) \quad (4.20)$$

$$= \frac{2\pi i}{L^2} \text{Tr}[PxP, PyP] \quad (4.21)$$

$$= \frac{2\pi i}{L^2} \text{Tr}P[[x, P], [y, P]] \quad (4.22)$$

In the limit of large L , we expect that the contribution to the trace is uniform in space, and so can replace the global trace with a local trace around some arbitrary point in the system $\mathbf{r}$, $\frac{1}{L^2} \text{Tr} \rightarrow \text{tr}_{\mathbf{r}}$. Thus we arrive at

$$\mathcal{Q}_{\text{Kit}} = 2\pi i \text{tr}_{\mathbf{r}}[P[x, P], [y, P]]. \quad (4.23)$$

Now let us construct the exact form of the projector onto a band for a translationally symmetric Hamiltonian. From Bloch's theorem, we know the projector will have the following form

$$P = \frac{1}{4\pi^2} \int d^2\mathbf{k} |k\rangle\langle k| \otimes \psi(\mathbf{k}) \quad (4.24)$$

This means that we can use the identity $[x, P] \rightarrow -i\partial_x\psi(\mathbf{k})$ to show that

$$\mathcal{Q}_{\text{Kit}} = \frac{-i}{2\pi} \int d^2\mathbf{k} \text{tr}_{\mathbf{r}}(|\mathbf{k}\rangle\langle\mathbf{k}| \otimes \psi(\mathbf{k})[\partial_x\psi(\mathbf{k}), \partial_y\psi(\mathbf{k})]) \quad (4.25)$$

Now, using the fact that $\langle\mathbf{r}|\mathbf{k}\rangle\langle\mathbf{k}|\mathbf{r}\rangle = 1$, this reduces to a trace over the unit cell degree of freedom only,

$$\mathcal{Q}_{\text{Kit}} = \frac{-i}{2\pi} \int d^2\mathbf{k} \text{Tr}(\psi(\mathbf{k})[\partial_x\psi(\mathbf{k}), \partial_y\psi(\mathbf{k})]). \quad (4.26)$$

Comparing to eqn. 3.33, we see that this quantity exactly evaluates an integral of the Berry curvature over the Brillouin zone. Thus we have shown that – at least in the context of an infinite, translationally symmetric system – $\mathcal{Q}_{\text{Kit}}$ evaluates the Chern number.

A Useful Identity for Operators in Momentum Space — Let us start by considering an arbitrary operator in a translationally symmetric system with the following form,

$$A = \int_{BZ} \frac{dk}{2\pi} |k\rangle\langle k| \otimes A(k), \quad (4.27)$$

where $A(k)$ acts on the degrees of freedom inside the unit cell. Here we use a 1D Brillouin zone, but this can be straightforwardly extended to higher dimensions. Let us consider the following expression,

$$[x, A] = \frac{1}{2\pi} \int_{BZ} dk [x, |k\rangle\langle k|] \otimes A(k). \quad (4.28)$$

using the identity $x|k\rangle = i\partial_k|k\rangle$, we can rewrite this in the form

$$[x, A] = \frac{1}{2\pi} \int_{BZ} dk i\partial_k (|k\rangle\langle k|) \otimes A(k) \quad (4.29)$$

$$= \frac{1}{2\pi} \int_{BZ} dk |k\rangle\langle k| \otimes (-i\partial_k A(k)), \quad (4.30)$$

where we used the fact that an integral of total derivative of the RHS vanishes, as the integral is over the whole Brillouin zone. Thus, we can make the general substitution

$$[x, A] \rightarrow -i\partial_k A(k) \quad (4.31)$$

4.1.2 Kitaev's Marker in a Finite System

So far, we have defined the marker for a system living in infinite boundary conditions. This version of the Chern number is useful in analytical arguments – for example, such a Chern number was used to prove the quantisation of Hall conductivity in the presence of disorder [19]. Generally, to be of practical use, this marker must be

4.2 The Bott Index

We will work with a lattice model, where unit cells are indexed by a vector of the form (m_x, m_y) . The lattice has size (L_x, L_y) , and we have set periodic boundary conditions. Sites are indexed by

$$|\mathbf{m}\alpha\rangle = |\mathbf{m}\rangle \otimes |\alpha\rangle, \quad (4.32)$$

where $\mathbf{m}$ is a vector labelling the unit cell and α selects a site within the unit cell. We shall use a Hamiltonian that has translational symmetry, so we may apply Bloch's theorem to obtain the form of the energy eigenstates,

$$\hat{H} = \sum_{\mathbf{k}} |\mathbf{k}\rangle \langle \mathbf{k}| \otimes \hat{H}(\mathbf{k}) \quad (4.33)$$

$$|\psi_{\mathbf{k},n}\rangle = |\mathbf{k}\rangle \otimes |u_{\mathbf{k},n}\rangle, \quad (4.34)$$

where $|u_{\mathbf{k},n}\rangle$ are the eigenstates of $\hat{H}(\mathbf{k})$ and the momentum states are defined as in eqn. 3.3,

$$|\mathbf{k}\rangle = \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{m}} e^{i\mathbf{m}\cdot\mathbf{k}} |\mathbf{m}\rangle, \quad (4.35)$$

and we assume that the system has a total of η bands, so $n \in \{1, \dots, \eta\}$. Henceforth we will assume that only the n^{th} band is occupied, so we can construct a projector onto the occupied subspace, $\hat{P}$, and its complement, $\hat{Q}$.

$$\hat{P} = \sum_{\mathbf{k}} |\psi_{\mathbf{k},n}\rangle \langle \psi_{\mathbf{k},n}| \quad (4.36)$$

$$\hat{Q} = \sum_{\mathbf{k}, m \neq n} |\psi_{\mathbf{k},m}\rangle \langle \psi_{\mathbf{k},m}|. \quad (4.37)$$

We also introduce a pair of operators $e^{i\delta_x \hat{x}}$ and $e^{i\delta_y \hat{y}}$, with

$$\delta_x = \frac{2\pi}{L_x}, \quad \delta_y = \frac{2\pi}{L_y}. \quad (4.38)$$

In order to understand the effect of these operators, we consider their effect on the plane wave states

$$e^{i\delta_x \hat{x}} |\mathbf{k}\rangle = \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{m}} e^{i\delta_x \hat{x}} e^{i\mathbf{m} \cdot \mathbf{k}} |\mathbf{m}\rangle \quad (4.39)$$

$$= \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{m}} e^{i\mathbf{m} \cdot (\mathbf{k} + \boldsymbol{\delta}_x)} |\mathbf{m}\rangle \quad (4.40)$$

$$= |\mathbf{k} + \boldsymbol{\delta}_x\rangle, \quad (4.41)$$

similarly

$$e^{i\delta_y \hat{y}} |\mathbf{k}\rangle = |\mathbf{k} + \boldsymbol{\delta}_y\rangle \quad (4.42)$$

Here the symbols $\boldsymbol{\delta}_x$ and $\boldsymbol{\delta}_y$ represents a vector translation in $\mathbf{k}$ -space,

$$\boldsymbol{\delta}_x = \begin{pmatrix} \delta_x \\ 0 \end{pmatrix}, \quad \boldsymbol{\delta}_y = \begin{pmatrix} 0 \\ \delta_y \end{pmatrix}. \quad (4.43)$$

Thus, these operators effect a translation in $\mathbf{k}$ space by the minimum step in the reciprocal lattice.

Now let's look at the effect of these operators on the projector $\hat{P}$,

$$e^{i\delta_x \hat{x}} \hat{P} e^{-i\delta_x \hat{x}} = \sum_{\mathbf{k}} e^{i\delta_x \hat{x}} |\mathbf{k}\rangle \langle \mathbf{k}| e^{-i\delta_x \hat{x}} \otimes |u_{\mathbf{k},n}\rangle \langle u_{\mathbf{k},n}| \quad (4.44)$$

$$= \sum_{\mathbf{k}} |\mathbf{k} + \boldsymbol{\delta}_x\rangle \langle \mathbf{k} + \boldsymbol{\delta}_x| \otimes |u_{\mathbf{k},n}\rangle \langle u_{\mathbf{k},n}| \quad (4.45)$$

$$= \sum_{\mathbf{k}} |\mathbf{k}\rangle \langle \mathbf{k}| \otimes |u_{\mathbf{k}-\boldsymbol{\delta}_x,n}\rangle \langle u_{\mathbf{k}-\boldsymbol{\delta}_x,n}|, \quad (4.46)$$

where the last line follows from a redefinition of $\mathbf{k}$. This is allowed as we are summing over the whole Brillouin zone. For the sake of clarity later on, we will re-express the density operator $|u_{\mathbf{k},n}\rangle \langle u_{\mathbf{k},n}|$ in the following form

$$|u_{\mathbf{k},n}\rangle \langle u_{\mathbf{k},n}| = \hat{u}_{\mathbf{k},n}, \quad (4.47)$$

so we get

$$e^{i\delta_x \hat{x}} \hat{P} e^{-i\delta_x \hat{x}} = \sum_{\mathbf{k}} |\mathbf{k}\rangle \langle \mathbf{k}| \otimes \hat{u}_{\mathbf{k}-\boldsymbol{\delta}_x,n}. \quad (4.48)$$

Now we have enough tools to introduce the definition of the Bott index!

$$B(\mathbf{A}) = -\frac{N}{2\pi} \sum_{\alpha} \Im \langle \mathbf{A}\alpha | \log \left[\hat{U} \hat{V} \hat{U}^{\dagger} \hat{V}^{\dagger} + \hat{Q} \right] | \mathbf{A}\alpha \rangle \quad (4.49)$$

Here N denotes the total number of unit cells in the system and the log is a standard matrix

logarithm, defined as

$$\log(M) = \sum_{k=1}^{\infty} (-1)^{k+1} \frac{(M - \mathbb{1})^k}{k}. \quad (4.50)$$

The Bott index is a local indicator, $\mathbf{A}$ is a vector that selects which unit cell in the system we are looking at, and

$$\hat{U} = \hat{P} e^{i\delta_x \hat{x}} \hat{P} \quad (4.51)$$

$$\hat{V} = \hat{P} e^{i\delta_y \hat{y}} \hat{P}. \quad (4.52)$$

To make sense of this expression, let us focus on the $\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger$ part only, writing this out explicitly gives us

$$\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger = \hat{P} e^{i\delta_x \hat{x}} \hat{P} e^{i\delta_y \hat{y}} \hat{P} e^{-i\delta_x \hat{x}} \hat{P} e^{-i\delta_y \hat{y}} \hat{P}. \quad (4.53)$$

We now add in the identity in the form $e^{i\delta_x \hat{x}} e^{-i\delta_x \hat{x}}$, as well as the corresponding expression for y , and regroup the equation, taking advantage of the fact that $\hat{x}$ and $\hat{y}$ commute, to get

$$\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger = \hat{P} \left[e^{i\delta_x \hat{x}} \hat{P} e^{-i\delta_x \hat{x}} \right] \left[e^{i\delta_y \hat{y}} e^{i\delta_x \hat{x}} \hat{P} e^{-i\delta_y \hat{y}} e^{-i\delta_x \hat{x}} \right] \left[e^{i\delta_y \hat{y}} \hat{P} e^{-i\delta_y \hat{y}} \right] \hat{P}. \quad (4.54)$$

Using eqn. 4.48 we can now rewrite this in terms of Bloch states to get

$$\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger = \sum_{\mathbf{k}} |\mathbf{k}\rangle \langle \mathbf{k}| \otimes \hat{u}_{\mathbf{k},n} \hat{u}_{\mathbf{k}-\delta_x,n} \hat{u}_{\mathbf{k}-\delta_x-\delta_y,n} \hat{u}_{\mathbf{k}-\delta_y,n} \hat{u}_{\mathbf{k},n}. \quad (4.55)$$

Since the first and last $\hat{u}$ density operator in this expression are identical, we can rewrite this in terms of a trace to get

$$\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger = \sum_{\mathbf{k}} |\mathbf{k}\rangle \langle \mathbf{k}| \otimes |u_{\mathbf{k},n}\rangle \langle u_{\mathbf{k},n}| \cdot \text{Tr} \left[\hat{u}_{\mathbf{k},n} \hat{u}_{\mathbf{k}-\delta_x,n} \hat{u}_{\mathbf{k}-\delta_x-\delta_y,n} \hat{u}_{\mathbf{k}-\delta_y,n} \right]. \quad (4.56)$$

If we compare this to the definition of Berry flux (eqn. 3.12), we see that this reduces to

$$\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger = \sum_{\mathbf{k}} |\mathbf{k}\rangle \langle \mathbf{k}| \otimes |u_{\mathbf{k},n}\rangle \langle u_{\mathbf{k},n}| \cdot r_{\mathbf{k}} e^{-iF_{\mathbf{k}} - \delta_x - \delta_y}, \quad (4.57)$$

where $r_{\mathbf{k}}$ is just some real number that we don't care about, arising from the u states not being perfectly parallel to one another. This is promising, remember we are aiming to get a sum of $F_{\mathbf{k}}$ over the whole Brillouin zone. Thus in order to extract the term in the exponential we would like to take a log. However, we cannot simply log the expression in the form we have here. This is because it only has support on the occupied subspace,

$$\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger \sim \begin{pmatrix} M & 0 \\ 0 & 0 \end{pmatrix}, \quad (4.58)$$

where M just indicates the presence of a nonzero operator. The presence of a null space means

a log of this matrix will diverge. To remedy this, we add in the $\hat{Q}$ projector,

$$\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger + \hat{Q} \sim \begin{pmatrix} M & 0 \\ 0 & \mathbb{1} \end{pmatrix}. \quad (4.59)$$

Now we can safely take the log of this matrix, getting the following result

$$\log \left[\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger + \hat{Q} \right] = \sum_{\mathbf{k}} |\mathbf{k}\rangle \langle \mathbf{k}| \otimes |u_{\mathbf{k},n}\rangle \langle u_{\mathbf{k},n}| \cdot (\log(r_{\mathbf{k}}) - iF_{\mathbf{k}-\delta_x-\delta_y}), \quad (4.60)$$

Next we trace out the internal degree of freedom and squeeze this expression between a pair of position states $|\mathbf{A}\rangle$, selecting some arbitrary unit cell in our system.

$$\sum_{\alpha} \langle \mathbf{A}\alpha | \log \left[\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger + \hat{Q} \right] | \mathbf{A}\alpha \rangle = \sum_{\mathbf{k}} |\langle \mathbf{A} | \mathbf{k} \rangle|^2 \cdot \langle u_{\mathbf{k},n} | u_{\mathbf{k},n} \rangle \cdot (\log(r_{\mathbf{k}}) - iF_{\mathbf{k}-\delta_x-\delta_y}). \quad (4.61)$$

We can show that $|\langle \mathbf{A} | \mathbf{k} \rangle|^2 = \frac{1}{N}$ and $\langle u_{\mathbf{k},n} | u_{\mathbf{k},n} \rangle = 1$. All that is left to do is take the imaginary part,

$$\sum_{\alpha} \Im \langle \mathbf{A}\alpha | \log \left[\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger + \hat{Q} \right] | \mathbf{A}\alpha \rangle = \frac{1}{N} \sum_{\mathbf{k}} -F_{\mathbf{k}-\delta_x-\delta_y}. \quad (4.62)$$

Now, redefining $\mathbf{k}$ and multiplying by $-\frac{N}{2\pi}$ gives us

$$-\frac{N}{2\pi} \sum_{\alpha} \Im \langle \mathbf{A}\alpha | \log \left[\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger + \hat{Q} \right] | \mathbf{A}\alpha \rangle = \frac{1}{2\pi} \sum_{\mathbf{k}} F_{\mathbf{k}}. \quad (4.63)$$

By comparing with eqn. 3.15, we can see that we have arrived at an expression for the Chern number!

$$\mathcal{Q} = -\frac{N}{2\pi} \sum_{\alpha} \Im \langle \mathbf{A}\alpha | \log \left[\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger + \hat{Q} \right] | \mathbf{A}\alpha \rangle. \quad (4.64)$$

Thus we have shown that in the case of a translationally symmetric system with periodic boundaries, the Bott index evaluated at any point in our material always exactly calculates the Chern number. Before we show some example calculations, let us introduce one more quantity.

4.3 Chern Marker

The Chern marker was first introduced in [25], however we will not restate the derivation that they used. Instead we will justify the Chern marker by showing that it can be obtained from the Bott index in the limit of large system size.

Our starting point is the expression for Bott index, eqn. 4.49. We start by multiplying $\hat{U}$ and $\hat{V}$ by complex phases $e^{-i\delta_x C_x}$ and $e^{-i\delta_y C_y}$. This has no effect on the value of the Bott index, since the phases have opposite sign in $\hat{U}^\dagger$ and $\hat{V}^\dagger$.

$$\hat{U} \rightarrow \hat{P} e^{i\delta_x(\hat{x}-C_x)} \hat{P} \quad (4.65)$$

$$\hat{V} \rightarrow \hat{P} e^{i\delta_y(\hat{y}-C_y)} \hat{P}. \quad (4.66)$$

As the system size becomes large, the factors δ_x and δ_y will become infinitesimally small. Around the neighbourhood of the unit cell at position (C_x, C_y) , the term in the exponential will be small so we can Taylor expand it,

$$\hat{U} = \hat{P} + i\delta_x \hat{P} \hat{x}_C \hat{P} + O(\hat{x}_C^2) \quad (4.67)$$

$$\hat{V} = \hat{P} + i\delta_y \hat{P} \hat{y}_C \hat{P} + O(\hat{y}_C^2) \quad (4.68)$$

where we have relabelled $\hat{x} - C_x = \hat{x}_C$ and $\hat{y} - C_y = \hat{y}_C$. Using these expressions, we can re-express the operator $\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger$ up to second order in δ as¹

$$\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger = \hat{P} - \delta_x \delta_y \left(\hat{P} \hat{x}_C \hat{P} \hat{y}_C \hat{P} - \hat{P} \hat{y}_C \hat{P} \hat{x}_C \hat{P} \right) + O(\delta_x^2, \delta_y^2) \quad (4.69)$$

Now we use the identities

$$\hat{P} + \hat{Q} = \mathbb{1} \quad (4.70)$$

$$\log[\mathbb{1} + \delta M] \simeq M, \text{ for } \delta \text{ small} \quad (4.71)$$

to show that

$$\log \left[\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger + Q \right] \simeq -\delta_x \delta_y \left[\hat{P} \hat{x}_C \hat{P}, \hat{P} \hat{y}_C \hat{P} \right] \quad (4.72)$$

We can now drop the constant shifts in $\hat{x}_C$ and $\hat{y}_C$, since these vanish due to the commutator. Inserting this into the expression for Bott index, we arrive at the form of the Chern marker,

$$C(\mathbf{A}) = -2\pi i \sum_{\alpha} \langle \mathbf{A}\alpha | \left[\hat{P} \hat{x} \hat{P}, \hat{P} \hat{y} \hat{P} \right] | \mathbf{A}\alpha \rangle. \quad (4.73)$$

We have skipped taking the imaginary part by noticing that the commutator is anti-Hermitian, so all of its eigenvalues are imaginary anyway, allowing us to make the substitution $\Im \rightarrow -i$.

This calculation comes with one important caveat. The Chern marker has an extra property that the Bott index lacks. If the system in question has finite size, then the sum of the Chern number over the whole system vanishes. This is because the sum of $C(\mathbf{A})$ over the system is given by

$$\sum_{\mathbf{A}} C(\mathbf{A}) = -2\pi i \text{Tr} \left[\hat{P} \hat{x} \hat{P}, \hat{P} \hat{y} \hat{P} \right], \quad (4.74)$$

and the trace of a commutator is always zero in a finite Hilbert space.

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¹You might have noticed that we have omitted the δ_x^2 and δ_y^2 terms. This is justified because those terms involve only hermitian operators, such as $\hat{P} \hat{x}_C \hat{P} \hat{x}_C \hat{P}$, and these give no contribution when we take the imaginary part later.

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Chapter 5

The Crosshair Marker

Chapter 6

Kitaev's Honeycomb Model

Chapter 7

The Amorphous Kitaev Model

Chapter 8

The Star Lattice Kitaev-Heisenberg Model

Chapter 9

Conclusion and Outlook